GNSS Receivers/Skydel Signal Simulation & Testing

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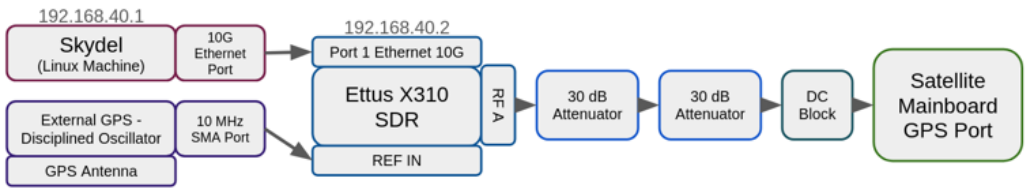
For anyone interested in validating that the GNSS receiver hardware will work at the expected altitude and velocity, the following GNSS simulation setup is described.

Table of Contents:

1. Overview
2. Skytraq Receivers
3. Skydel
4. SDR/Ettus X310 & Hardware
5. Simulation & Validation
6. Extra Resources

Overview:

In essence, a GNSS receiver expects a set of very precise radio-frequency signals from various satellites to determine location, velocity (depending on the receiver), and timing. To simulate these signals, we are using the following configuration:



GNSS Simulation to Mainboard Flowchart

Skytraq Receivers:

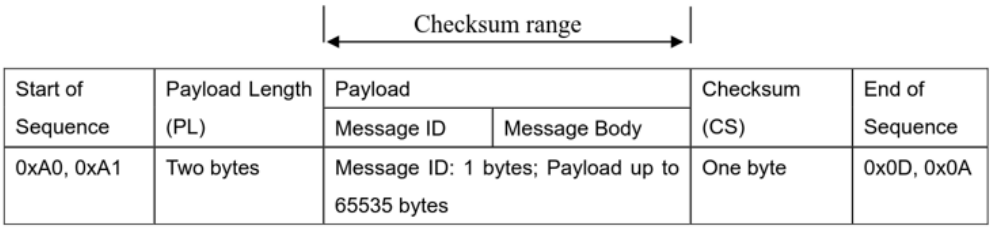
ARGUS will be flown with an S1216F8-GL Skytraq GNSS receiver, which is 'unlocked' for high altitudes and velocities.

However, depending on the mainboard, different receivers are used, as unlocked modules are significantly more expensive and in limited supply.

The two modules are the aforementioned S1216F8-GL and the PX1120S. Each has a different message protocol and message information. Shared between them are the relevant ECEF position and velocity frames, along with timing information.

> For reference, the **S1216F8-GL uses the AN0030 protocol** while the newer **PX1120S uses the AN0037 protocol**.

Both protocols have the following structure:



SkyTraq Technology, Inc.

- The checksum byte is simply the XOR of all preceding bytes in the *Payload* section of the message.

The S1216F8-GL periodically outputs a *0xDF* message that reports the receiver's navigation status, while the PX1120S outputs a *0xA8* message.

These two messages are slightly different. The information in the messages differs slightly, and the PX1120S units are on a 1/100 metric scale. See the AN0030 and AN0037 manuals (respectively) for greater details.

> For receivers that have not been previously configured, ensure that NMEA is disabled and binary message types are enabled.

Skydel:

Skydel is SAFRAN's GNSS simulation software that can generate the signals which a GNSS receiver would expect to see in orbit.

- Skydel requires a license that is machine-specific, ours is set to expire in 2027.

- For real-time simulation, an RTX 3070 or higher is recommended

There is a basic introductory course on how to use Skydel from SAFRAN that explains how to use the software. This takes ~1 hour to complete.

SDR & Hardware:

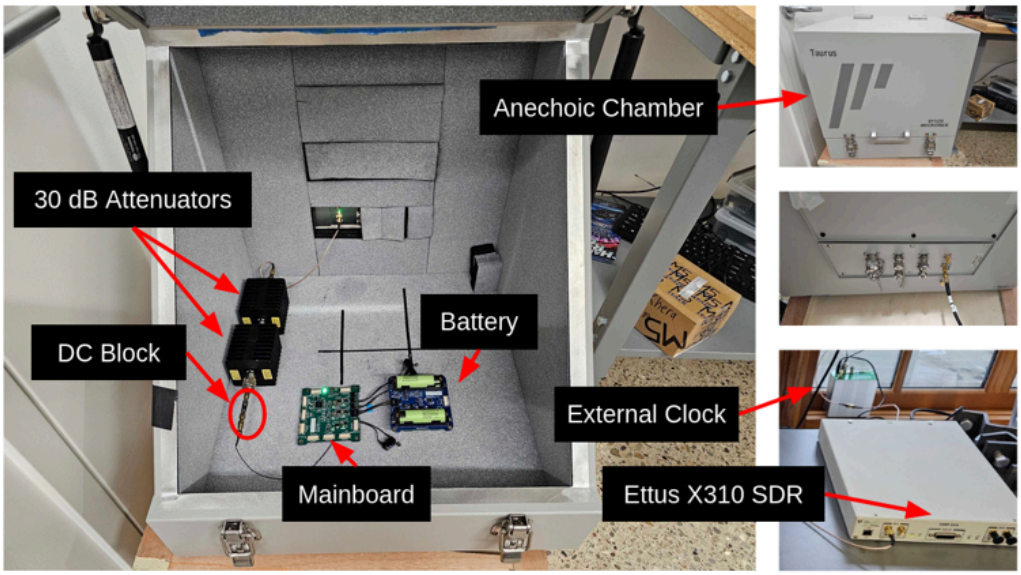
The Ettus X310 (not X300) is a **software-defined-radio** that converts the digital signal from Skydel to the necessary in-phase and quadrature components of the analog GNSS signals.

- The signals coming from the computer necessitate a 10G Ethernet (CAT 7 or greater) to facilitate enough bandwidth.
- The X310 requires a specific firmware that SAFRAN offers. Uploading the correct firmware is described thoroughly in their **user guide**. It is PERTINENT that you follow the guide CLOSELY. There are multiple ways to brick the X310 (which costs >$12,000), as well as mess up the dependencies needed for proper connection.
- An external 10MHz reference clock is required for signal generation. Previously, we have used a BG7TBL GPSDO.
- The Ethernet connection must be on PORT 1, the reference clock in REF IN, and the output RF A.

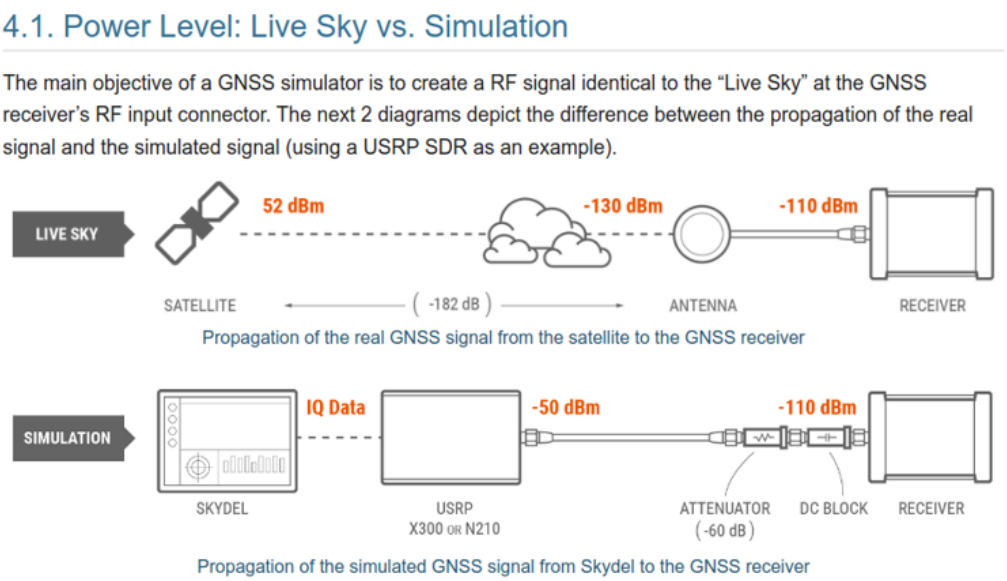
Two 30dB attenuators are placed in series after the X310 output to decrease the signal power to the average power of a true GNSS signal.

A DC Block is placed at the very end to prevent current from returning to the X310 and possibly damaging it.

- All of this is documented in the Skydel certification course and user manual.



⚠ NOTE: The mainboard is inside an anechoic/faraday chamber to prevent the possible leakage of simulated RF GNSS signals (which is very illegal). It was found that **without a sufficient enclosure, the signal will leak.**



From the SAFRAN Skydel Manual

Simulation & Validation:

Work in Progress ....

Extra Resources:

For anyone interested in learning more about GNSS, software-defined-radios, or related topics, here are some following resources:

The Stanford GPS Course: GPS: An Introduction to Satellite Navigation

- Fantastic and brief course on how GPS and satellite navigation works

GNU Radio Tutorials: https://wiki.gnuradio.org/index.php/Tutorials

- GNU Radio is the most widely used SDR toolkit and has plenty of good tutorials on the fundamentals

GNSS SDR: GNSS-SDR

- Built on GNU Radio, a good tool to both learn and experiment with how GNSS receivers actually work